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To the AWS Data Services Engineering Team,

AWS Data Services operates some of the most widely used database, storage, and analytics infrastructure in the world. The engineering challenges here—distributed query processing, petabyte-scale storage engines, fault-tolerant data platforms serving millions of customers—are the class of problems I want to work on as a new graduate. I bring hands-on production experience in distributed systems, data pipelines, cloud infrastructure on AWS and GCP, and a track record of building software that runs reliably at scale.

- **Distributed Systems and Production Ownership:** At **Isomer AI**, I shipped 10+ production AI features—multi-model LLM pipelines, Temporal.io agentic workflows, Auth0 RBAC—managing CI/CD, on-call incident resolution, and SLA commitments across enterprise environments. I built **CortexQ**, a Kubernetes-native distributed inference platform with KEDA autoscaling, circuit breaker logic, EWMA load balancing, and OpenTelemetry observability, load tested to 500 concurrent users with 88 integration tests validating correctness.
- **Data Pipelines and Analytics Systems:** At **Capgemini**, I built distributed Kafka + Spark + Hadoop pipelines for real-time ingestion and batch processing that cut ticket-processing latency by 20%. My **PulseRank** project architected a multi-DB system—MySQL transactional writes, MongoDB feed projections with TTL indexes, Redis sorted sets for sub-millisecond leaderboard reads—validated for consistency under 10K concurrent users with Testcontainers.
- **AWS and Cloud-Native Infrastructure:** At **Mphasis**, I automated AWS and GCP CI/CD deployments and integrated Grafana dashboards for real-time production monitoring. My tech stack spans PostgreSQL, Redis, DuckDB, Docker, Kubernetes, and Helm—the same infrastructure AWS Data Services is built on.
- **GenAI and ML Systems:** I built a vendor-neutral **LLM Evaluation Harness** tracking hallucination rates, p50/p95 latency, and cost-per-call across Claude, GPT-4o, and Gemini on 2,000 labeled examples—applying McNemar significance tests and Bootstrap confidence intervals. At NC State, I deployed a genomic data platform enabling 100+ researchers to analyze 500GB+ datasets, improving ML model accuracy by 12%.

AWS Data Services is foundational infrastructure for the rest of the cloud. That is precisely the kind of high-ownership, high-impact engineering I want to build my career on. I would welcome the opportunity to discuss how my background fits your team's needs.

Sincerely,

Aditya Kulkarni